

Build a Simple Inventory Management System

Objective:

Create a RESTful API for managing an inventory of products. The API should support basic CRUD operations and include some business logic for managing stock levels and creating orders.

Requirements:

- Setup:
 1. Use Node.js and Express.js to build the API.
 2. Use a database (e.g., lowdb, mongodb) for data persistence.
 3. Use CQRS pattern.
 4. Include appropriate environment configuration using .env files if necessary.
- API:
 1. Products:
 - a. Add new endpoint
GET /products: Retrieve a list of all products.
 - b. Add new endpoint
POST /products: Create a new product (fields: name, description, price, stock; all required, max length 50).
 2. Stock Management:
 - a. Add a new endpoint
POST /products/:id/restock: to increase the stock level of a product.
 - b. Add a new endpoint
POST /products/:id/sell: to decrease the stock level of a product. Ensure the stock cannot go below zero.
 3. Price Validation:
 - a. Ensure that the price of a product is always positive.
 4. Order Management:
 - a. Add endpoints to manage orders:
POST /orders: Create a new order (fields: customerId, products).
 - b. Implement logic to update stock levels when an order is placed.
 - c. Prevent orders from being placed if stock is insufficient.

- **Validation:**

1. Implement input validation for all endpoints using a library like Joi or express-validator.

- **Error Handling:**

1. Implement proper error handling for cases such as invalid input, resource not found, and server errors.
2. Return appropriate HTTP status codes and error messages.

- **Submission Guidelines:**

1. Provide a link to a GitHub repository containing the code with pr to main branch.
2. Ensure the project is well-structured and follows best practices for Node.js development.